

Note	Definition
1	The Processor Base Power (a.k.a TDP) and Assured Power (cTDP) values are the average power dissipation in junction temperature operating condition limit, for the SKU Segment and Configuration, for which the processor is validated during manufacturing when executing an associated Intel-specified high-complexity workload at the processor IA core frequency corresponding to the configuration and SKU.
2	Thermal workload (Processor Base Power (a.k.a TDP)) may consist of a combination of processor IA core intensive and graphics core intensive applications.
3	Can be modified at runtime by MSR writes, with MMIO and with PECI commands.
4	'Turbo Time Parameter' is a mathematical parameter (units of seconds) that controls the processor turbo algorithm using a moving average of energy usage. Do not set the Turbo Time Parameter to a value less than 0.1 seconds. refer to Platform Power Control on page 74 for further information.
5	The shown limit is a time averaged-power, based upon the Turbo Time Parameter. Absolute product power may exceed the set limits for short durations or under virus or uncharacterized workloads.
6	The Processor will be controlled to a specified power limit as described in Intel® Turbo Boost Technology 2.0 Power Monitoring on page 45. If the power value and/or 'Turbo Time Parameter' is changed during runtime, it may take a short period of time (approximately 3 to 5 times the 'Turbo Time Parameter') for the algorithm to settle at the new control limits.
7	This is a hardware default setting and not a behavioral characteristic of the part.
8	For controllable turbo workloads, the PL2 limit may be exceeded for up to 10ms.
9	LPM power level is an opportunistic power and is not a guaranteed value as usages and implementations may vary.
10	Power limits may vary depending on if the product supports the Minimum Assured Power (cTDP Down) and/or Maximum Assured Power (cTDP Up) modes. Default power limits can be found in the PKG_PWR_SKU MSR (614h).
11	The processor die do not reach maximum sustained power simultaneously since the sum of the 2 die's estimated power budget is controlled to be equal to or less than the package Processor Base Power (a.k.a TDP) (PL1) limit.
12	Minimum Assured Power(cTDP Down) power is based on 96EU equivalent graphics configuration. Minimum Assured Power(cTDP Down) does not decrease the number of active Processor Graphics EUs but relies on Power Budget Management (PL1) to achieve the specified power level.
13	May vary based on SKU.
14	- The formula of $PL2 = PL1 * 1.25$ is the hardware. - PL2- SoC opportunistic higher Average Power with limited duration controlled by Tau_PL1 setting, the larger the Tau, the longer the PL2 duration. - A recommendation to set all power delivery especially PL2 and PL1 based on platform power and thermal capability via BIOS.
15	Possessor Base Power (a.k.a TDP) workload does not reflect various I/O connectivity cases such as Thunderbolt.
16	Hardware default of PL1 Tau=1s, By including the benefits available from power and thermal management features the recommended is to use PL1 Tau=28s.

4.2.1 Processor Line Power and Frequency Specifications

Table 12. Processor Base Power (TDP) and Frequency Specifications(H-Processor Line, P-Processor Line)

Segment and Package	Processor IA/GT Cores, and Processor Base Power	Configuration	Processor IA Core Frequency	Graphics Core Frequency	Processor Base Power [w]	Notes
H-Processor Line BGA	6+8 Core 45W	Maximum Assured Power	2.7GHz up to 2.9GHz	300MHz	65	1,9,10, 11,12, 15
		P-Core	2.3GHz up to 2.5GHz		45	
		E-Core	1.7GHz up to 1.8GHz		35	
		Minimum Assured Power	1.5GHz up to 1.9GHz			
		LFM	400MHZ	100MHz	N/A	
H-Processor Line BGA	6+4 Core 45W	Maximum Assured Power	2.7GHz	300MHz	65	1,9,10, 11,12, 15
		P-Core	2.3GHz		45	
		E-Core	1.7GHz		35	
		Minimum Assured Power	1.5GHz			
		LFM	400MHZ	100MHz	N/A	
H-Processor Line BGA	4+8 Core 45W	Maximum Assured Power	3.1GHz up to 3.2GHz	300MHz	65	1,9,10, 11,12, 15
		P-Core	2.5GHz up to 2.7GHz		45	
		E-Core	1.8GHz up to 2.0GHz		35	
		Minimum Assured Power	1.7GHz up to 2.2GHz			
		LFM	400MHZ	100MHz	N/A	
H-Processor Line BGA	4+4 Core 45W	Maximum Assured Power	2.5GHz	300MHz	65	1,9,10, 11,12, 15
		P-Core	2.0GHz		45	
		E-Core	1.5GHz		35	
		Minimum Assured Power	1.1GHz			
		LFM	400MHZ	100MHz	N/A	
P-Processor Line BGA	6+8 Core 28W	Maximum Assured Power	2GHz	300MHz	35	1,9,10, 11,12, 15
		P-Core	1.8GHz		28	
		E-Core	1.3GHz		20	
		Minimum Assured Power	1.4GHz			

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Segment and Package	Processor IA/GT Cores, and Processor Base Power	Configuration	Processor IA Core Frequency	Graphics Core Frequency	Processor Base Power [w]	Notes
P-Processor Line BGA	4+8 Core 28W	LFM	400MHZ	100MHz	N/A	1,9,10, 11,12, 15
		Maximum Assured Power	2.1GHz up to 2.5GHz	300MHz	35	
		P-Core	1.7GHz up to 2.2GHz		28	
		E-Core	1.2GHz up to 1.6GHz		20	
		Minimum Assured Power	1.4GHz up to 1.6GHz			
P-Processor Line BGA	2+8 Core 28W	LFM	400MHZ	100MHz	N/A	1,9,10, 11,12, 15
		Maximum Assured Power	2GHz	300MHz	35	
		P-Core	1.5GHz		28	
		E-Core	1.1Ghz		20	
		Minimum Assured Power	1.0GHz			

Table 13. Processor Base Power (TDP) and Frequency Specifications(U-Processor Line)

Segment and Package	Processor IA/GT Cores, and Processor Base Power	Configuration	Processor IA Core Frequency	Graphics Core Frequency	Processor Base Power [w]	Notes
U-Processor Line BGA	2+8 Core 15W	Maximum Assured Power	2.5GHz up to 2.7GHz	300MHz	28	1,9,10, 11,12, 15
		P-Core	1.3GHz up to 1.8GHz		15	
		E-Core	0.9GHz up to 1.3GHz		12	
		Minimum Assured Power	0.8GHz up to 1.3GHz			
		LFM	400MHZ	100MHz	N/A	
U-Processor Line BGA	2+4 Core 15W	Maximum Assured Power	2.5GHz	300MHz	28	1,9,10, 11,12, 15
		P-Core	1.2GHz		15	
		E-Core	0.9GHz		12	
		Minimum Assured Power	0.8GHz			
		LFM	400MHZ	100MHz	N/A	
U-Processor Line BGA	1+4 Core 15W	Maximum Assured Power	2.5GHz	300MHz	28	1,9,10, 11,12, 15
		P-Core	1.1GHz up to 1.2GHz		15	

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Segment and Package	Processor IA/GT Cores, and Processor Base Power	Configuration	Processor IA Core Frequency	Graphics Core Frequency	Processor Base Power [w]	Notes
		E-Core	0.9GHz		12	
		Minimum Assured Power	0.8GHz up to 0.9GHz			
		LFM	400MHZ	100MHZ	N/A	

Table 14. Processor Base Power (TDP) and Frequency Specifications (U 9W-Processor Line)

Segment and Package	Processor IA /GT Cores, and Processor Base Power	Configuration	Processor IA Core Frequency	Graphics Core Frequency	Processor Base Power [w]	Notes
U-Processor Line BGA	2+8 Core 9W	Maximum Assured Power	1.7GHz up to 1.9GHz	300MHz	15	1,9,10, 11,12, 15
		P-Core	1.0GHz up to 1.1GHz		9	
		E-Core	0.7GHz up to 0.8GHz			
		LFM	400MHZ	100MHZ	N/A	
U-Processor Line BGA	2+4 Core 9W	Maximum Assured Power	1.7GHz	300MHz	15	1,9,10, 11,12, 15
		P-Core	1.0GHz		9	
		E-Core	0.7GHz			
		LFM	400MHZ	100MHZ	N/A	
U-Processor Line BGA	1+4 Core 9W	Maximum Assured Power	1.6GHz	300MHz	15	1,9,10, 11,12, 15
		P-Core	1.0GHz		9	
		E-Core	0.7GHz			
		LFM	400MHZ	100MHZ	N/A	

Table 15. Processor Base Power (TDP) and Frequency Specifications (S-Processor Line)

Segment and Package	Processor IA/GT Cores, and Processor Base Power	Configuration	Processor IA Core Frequency	Graphics Core Frequency	Processor Base Power [w]	Notes
S-Processor Line LGA	8+8 Core 150W	P-Core	3.4 GHz	300MHZ	150	1,9,10, 11,12, 15
		E-Core	2.5 GHz			
		LFM	400MHZ	300MHZ	N/A	
S-Processor Line LGA	8+8 Core 125W	P-Core	3.2GHz	300MHZ	125	1,9,10, 11,12, 15
		E-Core	2.4GHz			
		LFM	400MHZ	300MHZ	N/A	

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Segment and Package	Processor IA/GT Cores, and Processor Base Power	Configuration	Processor IA Core Frequency	Graphics Core Frequency	Processor Base Power [w]	Notes
S-Processor Line LGA	8+4 Core 125W	P-Core	3.6GHz	300MHZ	125	1,9,10, 11,12, 15
		E-Core	2.7GHz			
		LFM	400MHZ	300MHZ	N/A	
S-Processor Line LGA	6+4 Core 125W	P-Core	3.7GHz	300MHZ	125	1,9,10, 11,12, 15
		E-Core	2.8GHz			
		LFM	400MHZ	300MHZ	N/A	
S-Processor Line LGA	8+8 Core 65W	P-Core	2.4GHz	300MHZ	65	1,9,10, 11,12, 15
		E-Core	1.8GHz			
		LFM	400MHZ		N/A	
S-Processor Line LGA	8+4 Core 65W	P-Core	2.1GHz	300MHZ	65	1,9,10, 11,12, 15
		E-Core	1.6GHz			
		LFM	400MHZ		N/A	
S-Processor Line LGA	8+8 Core 35W	P-Core	1.4GHz	300MHZ	35	1,9,10, 11,12, 15
		E-Core	1GHz			
		LFM	400MHZ		N/A	
S-Processor Line LGA	8+4 Core 35W	P-Core	1.4GHz	300MHZ	35	1,9,10, 11,12, 15
		E-Core	1GHz			
		LFM	400MHZ		N/A	
S-Processor Line LGA	6+0 Core 35W	P-Core	1.8GHz up to 2.1GHz	300MHZ	35	1,9,10, 11,12, 15
		LFM	1GHz		N/A	
S-Processor Line LGA	4+0 Core 35W	P-Core	2.2GHz up to 2.3GHz	300MHZ	35	1,9,10, 11,12, 15
		LFM	400MHZ		N/A	
S-Processor Line LGA	2+0 Core 35W	P-Core	2.8GHz up to 3.1GHz	300MHZ	35	1,9,10, 11,12, 15
		LFM	400MHZ		N/A	
S-Processor Line LGA	6+0 Core 65W	P-Core	2.5GHz up to 3.3GHz	300MHZ	65	1,9,10, 11,12, 15
		LFM	400MHZ		N/A	
S-Processor Line LGA	4+0 Core 60W	P-Core	3.3GHz up to 3.5GHz	300MHZ	60	1,9,10, 11,12, 15
		LFM	400MHZ		N/A	
S-Processor Line LGA	4+0 Core 58W	P-Core	3.3GHz	300MHZ	58	1,9,10, 11,12, 15

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Segment and Package	Processor IA/GT Cores, and Processor Base Power	Configuration	Processor IA Core Frequency	Graphics Core Frequency	Processor Base Power [w]	Notes
		LFM	400MHZ		N/A	
S-Processor Line LGA	2+0 Core 46W	P-Core	3.4GHz up to 3.7GHz	300MHZ	46	1,9,10, 11,12, 15
		LFM	400MHZ		N/A	

Table 16. Processor Base Power (TDP) and Frequency Specifications (HX-Processor Line)

Segment and Package	Processor IA/GT Cores, and Processor Base Power	Configuration	Processor IA Core Frequency	Graphics Core Frequency	Processor Base Power [w]	Notes
HX-Processor Line SBGA	8+8 Core 55W	P-Core	2 GHz up to 2.3 GHz	300 MHz	55 W	1,9,10, 11,12, 15
		E-Core	1.5 GHz up to 1.7 GHz		45 W	
		Minimum Assured Power (Configurable TDP-Down)	1.7 GHz up to 2 GHz			
		LFM	0.8 GHz		N/A	
HX-Processor Line SBGA	4+8 Core 55W	P-Core	2.5 GHz	300 MHz	55 W	
		E-Core	1.8 GHz		45 W	
		Minimum Assured Power (Configurable TDP-Down)	2.1 GHz			
		LFM	0.8 GHz		N/A	

4.2.2 Processor Line Thermal and Power

Table 17. Package Turbo Specifications (H/P/U -Processor Lines)

Segment and Package	Processor IA Cores, Graphics, Configuration and Processor Base Power (a.k.a. TDP)	Parameter	Minimum	Recommended Value	Tau MSR Max Value	Units	Notes
H-Processor Line	6+8 Core 45W	Power Limit 1 Time (PL1 Tau)	0.1	56	448	S	3,4,5,6, 7,8,14,1 6,17
		Power Limit 1 (PL1)	N/A	45	N/A	W	
		Power Limit 2 (PL2)	N/A	115	N/A	W	
H-Processor Line	6+4 Core 45W	Power Limit 1 Time (PL1 Tau)	0.1	56	448	S	3,4,5,6, 7,8,14,1 6,17

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Segment and Package	Processor IA Cores, Graphics, Configuration and Processor Base Power (a.k.a. TDP)	Parameter	Minimum	Recommended Value	Tau MSR Max Value	Units	Notes
		Power Limit 1 (PL1)	N/A	45	N/A	W	
		Power Limit 2 (PL2)	N/A	115	N/A	W	
H-Processor Line	4+8 Core 45W	Power Limit 1 Time (PL1 Tau)	0.1	56	448	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	45	N/A	W	
		Power Limit 2 (PL2)	N/A	95	N/A	W	
H-Processor Line	4+4 Core 45W	Power Limit 1 Time (PL1 Tau)	0.1	56	448	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	45	N/A	45	
		Power Limit 2 (PL2)	N/A	95	N/A	Note	
P-Processor Line	6+8 Core 28W	Power Limit 1 Time (PL1 Tau)	0.1	28	448	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	28	N/A	W	
		Power Limit 2 (PL2)	N/A	Note	N/A	W	
P-Processor Line	4+8 Core 28W	Power Limit 1 Time (PL1 Tau)	0.1	28	448	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	28	N/A	W	
		Power Limit 2 (PL2)	N/A	Note	N/A	W	
P-Processor Line	2+8 Core 28W	Power Limit 1 Time (PL1 Tau)	0.1	28	448	S	3,4,5,6,7,8,14,16,1
		Power Limit 1 (PL1)	N/A	28	N/A	W	
		Power Limit 2 (PL2)	N/A	Note	N/A	W	
U-Processor Line	2+8 Core 15W	Power Limit 1 Time (PL1 Tau)	0.1	28	448	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	15	N/A	W	
		Power Limit 2 (PL2)	N/A	Note	N/A	W	
U-Processor Line	2+4 Core 15W	Power Limit 1 Time (PL1 Tau)	0.1	28	448	S	3,4,5,6,7,8,14,16,1

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Segment and Package	Processor IA Cores, Graphics, Configuration and Processor Base Power (a.k.a. TDP)	Parameter	Minimum	Recommended Value	Tau MSR Max Value	Units	Notes
		Power Limit 1 (PL1)	N/A	15	N/A	W	
		Power Limit 2 (PL2)	N/A	Note	N/A	W	
U-Processor Line	1+4 Core 15W	Power Limit 1 Time (PL1 Tau)	0.1	28	448	S	
		Power Limit 1 (PL1)	N/A	15	N/A	W	3,4,5,6,7,8,14,16,1
		Power Limit 2 (PL2)	N/A	Note	N/A	W	
U-Processor Line	2+8 Core 9W	Power Limit 1 Time (PL1 Tau)	0.1	28	448	S	
		Power Limit 1 (PL1)	N/A	9	N/A	W	3,4,5,6,7,8,14,16,17
		Power Limit 2 (PL2)	N/A	Note	N/A	W	
U-Processor Line	2+4 Core 9W	Power Limit 1 Time (PL1 Tau)	0.1	28	448	S	
		Power Limit 1 (PL1)	N/A	9	N/A	W	3,4,5,6,7,8,14,16,1
		Power Limit 2 (PL2)	N/A	Note	N/A	W	
U-Processor Line	1+4 Core 9W	Power Limit 1 Time (PL1 Tau)	0.1	28	448	S	
		Power Limit 1 (PL1)	N/A	9	N/A	W	3,4,5,6,7,8,14,16,1
		Power Limit 2 (PL2)	N/A	Note	N/A	W	

Notes:

- No Specifications for Min/Max PL1/PL2 values.
- Hardware default of PL1 Tau=1s, By including the benefits available from power and thermal management features the recommended is to use PL1 Tau=28s.
- PL2- SoC opportunistic higher Average Power – Reactive, Limited Duration controlled by Tau_PL1 setting.
- PL1 Tau - PL1 average power is controlled via PID algorithm with this Tau, The larger the Tau, the longer the PL2 duration.
- System cooling solution and designs found to not being able to support the Performance Tau PL1, adjust the TauPL1 to cooling capability.
- A recommendation to set all power delivery especially PL2 and PL1 based on platform power and thermal capability via BIOS.

Table 18. Junction Temperature Specifications (H / HX / P/U - Processor Lines)

Segment	Symbol	Package Turbo Parameter	Temperature Range		Processor Base Power (a.k.a TDP) Specification Temperature Range		Units	Notes
			Minimum	Maximum	Minimum	Maximum		
HX-Processor Line SBGA	Tj	Junction temperature limit	0	100	0	100	°C	1, 3
H Processor Line BGA	Tj	Junction temperature limit	0	100	35	100	°C	1, 3
P/ U 15W - Processor Line BGA	Tj	Junction temperature limit	0	100	35	100	°C	1, 3
U 9W- Processor Line BGA	Tj	Junction temperature limit	0	100	35	90	°C	1, 2,3
Notes: 1. The thermal solution needs to ensure that the processor temperature does not exceed the Processor Base Power Specification Temperature. 2. For M - Processor line specification, thermal designs should ensure a Tjmax of 90C in sustained Processor Base Power (a.k.a TDP) workload for guaranteed performance. TCC Offset=10 and Tau value should be programed into MSR 1A2h. The recommended TCC_Offset averaging Tau is 5s. Operating the part above 90C will result in higher power. Refer to Turbo Implementation Guide (TIG) for evaluate TCC_Offset averaging Tau values. 3. The processor junction temperature is monitored by Digital Temperature Sensors (DTS). For DTS accuracy, refer to Digital Thermal Sensor on page 79.								

Table 19. Package Turbo Specifications (S / HX - Processor Lines)

Segment and Package	Processor IA Cores, Graphics, Configuration and Processor Base Power (a.k.a TDP)	Parameter	Minimum	Recommended Value	Tau MSR Max Value	Units	Notes
S-Processor Line LGA	8+8 Core 150W	Power Limit 1 Time (PL1 Tau)	0.1	56	448	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	150	N/A	W	
		Power Limit 2 (PL2)	N/A	241	N/A	W	
	8+8 Core 125W	Power Limit 1 Time (PL1 Tau)	0.1	56	448	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	125	N/A	W	
		Power Limit 2 (PL2)	N/A	241	N/A	W	
	8+4 Core 125W	Power Limit 1 Time (PL1 Tau)	0.1	56	448	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	125	N/A	W	
		Power Limit 2 (PL2)	N/A	190	N/A	W	
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Segment and Package	Processor IA Cores, Graphics, Configuration and Processor Base Power (a.k.a TDP)	Parameter	Minimum	Recommended Value	Tau MSR Max Value	Units	Notes
	6+4 Core 125W	Power Limit 1 Time (PL1 Tau)	0.1	56	448	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	125	N/A	W	
		Power Limit 2 (PL2)	N/A	150	N/A	W	
S-Processor Line LGA	8+8 Core 65W	Power Limit 1 Time (PL1 Tau)	0.1	28	448	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	65	N/A	W	
		Power Limit 2 (PL2)	N/A	202	N/A	W	
	8+4 Core 65W	Power Limit 1 Time (PL1 Tau)	0.1	28	448	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	65	N/A	W	
		Power Limit 2 (PL2)	N/A	180	N/A	W	
	8+8 Core 35W	Power Limit 1 Time (PL1 Tau)	0.1	28	448	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	35	N/A	W	
		Power Limit 2 (PL2)	N/A	106	N/A	W	
	8+4 Core 35W	Power Limit 1 Time (PL1 Tau)	0.1	28	448	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	35	N/A	W	
		Power Limit 2 (PL2)	N/A	99	N/A	W	
	6+0 Core 35W	Power Limit 1 Time (PL1 Tau)	0.1	28	448	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	35	N/A	W	
		Power Limit 2 (PL2)	N/A	74	N/A	W	
	4+0 Core 35W	Power Limit 1 Time (PL1 Tau)	0.1	28	448	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	35	N/A	W	
		Power Limit 2 (PL2)	N/A	69	N/A	W	
continued...							

Segment and Package	Processor IA Cores, Graphics, Configuration and Processor Base Power (a.k.a TDP)	Parameter	Minimum	Recommended Value	Tau MSR Max Value	Units	Notes
	2+0 Core 35W	Power Limit 1 Time (PL1 Tau)	0.1	28	448	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	35	N/A	W	
		Power Limit 2 (PL2)	N/A	35	N/A	W	
	2+0 Core 46W	Power Limit 1 Time (PL1 Tau)	0.1	28	448	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	46	N/A	W	
		Power Limit 2 (PL2)	N/A	46	N/A	W	
	6+0 Core 65W	Power Limit 1 Time (PL1 Tau)	0.1	28	448	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	65	N/A	W	
		Power Limit 2 (PL2)	N/A	117	N/A	W	
	4+0 Core 60W	Power Limit 1 Time (PL1 Tau)	0.1	28	448	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	60	N/A	W	
		Power Limit 2 (PL2)	N/A	89	N/A	W	
	4+0 Core 58W	Power Limit 1 Time (PL1 Tau)	0.1	28	448	S	3,4,5,6,7,8,14,16,17
		Power Limit 1 (PL1)	N/A	58	N/A	W	
		Power Limit 2 (PL2)	N/A	89	N/A	W	
HX-Processor Line SBGA	8+8 Core 55W	Power Limit 1 Time (PL1 Tau)	0.1	56	448	S	3,4,5,6,7,8,14,16
		Power Limit 1 (PL1)	N/A	55	N/A	W	
		Power Limit 2 (PL2)	N/A	Note	N/A	W	
HX-Processor Line SBGA	4+8 Core 55W	Power Limit 1 Time (PL1 Tau)	0.1	56	448	S	3,4,5,6,7,8,14,16

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Segment and Package	Processor IA Cores, Graphics, Configuration and Processor Base Power (a.k.a TDP)	Parameter	Minimum	Recommended Value	Tau MSR Max Value	Units	Notes
		Power Limit 1 (PL1)	N/A	55	N/A	W	
		Power Limit 2 (PL2)	N/A	Note	N/A	W	

Notes:

- No Specifications for Min/Max PL1/PL2 values.
- Hardware default of PL1 Tau=1s, By including the benefits available from power and thermal management features the recommended is to use PL1 Tau=28s.
- PL2- SoC opportunistic higher Average Power – Reactive, Limited Duration controlled by Tau_PL1 setting.
- PL1 Tau - PL1 average power is controlled via PID algorithm with this Tau. The larger the Tau, the longer the PL2 duration.
- System cooling solution and designs found to not being able to support the Performance TauPL1, adjust the TauPL1 to cooling capability.

Low Power and TTV

Table 20. Low Power and TTV Specifications (S-Processor Line LGA)

Processor IA Cores, Graphics Configuration and Processor Base Power (a.k.a TDP)	PCG7	Maximum Power Package C7 (W) ^{1,4,5}	Maximum Power Package C8 (W) ^{1,4,5}	TTV Processor Base Power (a.k.a TDP) (W) ^{6,7}	Min TCASE (°C)	Maximum TTV TCASE (°C)
8+8 Core 150W	2020E	N/A	N/A	150	0	59.2
8+8 Core 125W	2020A	N/A	N/A	125	0	61.9
8+4 Core 125W		N/A	N/A	125	0	61.9
6+4 Core 125W		N/A	N/A	125	0	61.9
8+8 Core 65W	2020C	N/A	N/A	65	0	71.3
8+4 Core 65W		N/A	N/A	65	0	71.3
6+0 -Core 65W		N/A	N/A	65	0	71.3 Note 9
4+0 Core 65W		N/A	N/A	65	0	71.3 Note 9
4+0 Core 60W		N/A	N/A	60	0	68.9
4+0 Core 58W		N/A	N/A	58	0	68.9
2+0 -Core 46W		N/A	N/A	46	0	62.2
8+8 Core 35W	2020D	N/A	N/A	35	0	65.5
8+4 Core 35W		N/A	N/A	35	0	65.5
6+0 -Core 35W		N/A	N/A	35	0	65.5

continued...

Processor IA Cores, Graphics Configuration and Processor Base Power (a.k.a TDP)	PCG7	Maximum Power Package C7 (W) ^{1,4,5}	Maximum Power Package C8 (W) ^{1,4,5}	TTV Processor Base Power (a.k.a TDP) (W) ^{6,7}	Min TCASE (°C)	Maximum TTV TCASE (°C)
4+0 -Core 35W		N/A	N/A	35	0	65.5
2+0 -Core 35W		N/A	N/A	35	0	65.5

Notes: 1. The package C-state power is the worst case power in the system configured as follows:
a. Memory configured for DDR4 and populated with two DIMMs per channel.
b. DMI and PCIe links are at L1
2. Specification at DTS = 50 °C and minimum voltage loadline.
3. Specification at DTS = 35 °C and minimum voltage loadline.
4. These DTS values in Notes 2 - 3 are based on the TCC Activation MSR having a value of 100, Refer [Thermal Management Features](#) on page 76.
5. These values are specified at VCC_MAX and VNOM for all other voltage rails for all processor frequencies. Systems should be designed to ensure the processor is not to be subjected to any static VCC and ICC combination wherein VCCP exceeds VCCP_MAX at specified ICCP. Refer the loadline specifications.
6. Thermal Processor Base Power (a.k.a TDP) should be used for processor thermal solution design targets. Processor Base Power is not the maximum power that the processor can dissipate. Processor Base Power (a.k.a TDP) is measured at DTS = -1. Processor Base Power(a.k.a TDP) is achieved with the Memory configured for DDR
7. Platform Compatibility Guide (PCG) (previously known as FMB) provides a design target for meeting all planned processor frequency requirements.
8. Not 100% tested. Specified by design characterization.

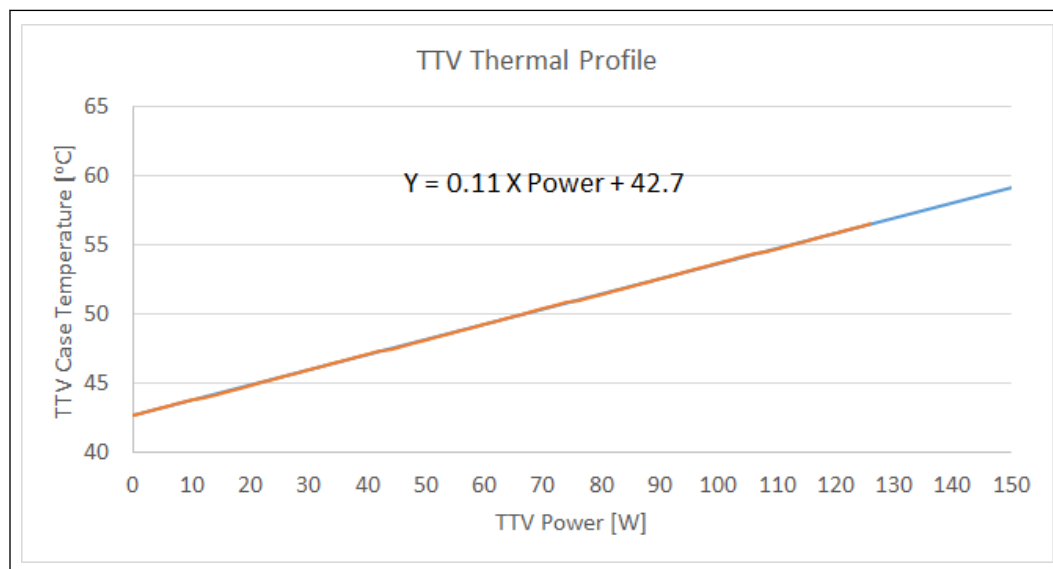
Table 21. TCONTROL Offset Configuration (S-Processor Line - Client)

Segment	8+8 Core	8+8 Core	8+8 Core		8+4 Core	6+4 Core	6+0 Core		4+0			2+0 Core	
Process or Base Power (a.k.a TDP) [W]	150	125	65	35	125	125	65	35	60	58	35	46	35
TEMP_TARGET (TCONTROL) [°C]	20	20	20	20	20	20	20	20	20	20	20	20	20
Notes:	• Digital Thermal Sensor (DTS) based fan speed control is recommended to achieve optimal thermal performance. • Intel recommends full cooling capability at approximately the DTS value of -1, to minimize TCC activation risk. • For example, if TCONTROL = 20 °C, Fan acceleration operation will start at 80 °C (100 °C - 20 °C).												

4.3 Thermal Profile for PCG 2020 Processor

4.3.1 Thermal Profile for PCG 2022E Processor

Figure 18. Thermal Profile for PCG 2022E Processor



NOTE

Refer to the following table for discrete points that constitute the thermal profile.

Table 22. Thermal Test Vehicle Thermal Profile for PCG 2022E Processor

Power (W)	T _{CASE_MAX} (°C)	Power (W)	T _{CASE_MAX} (°C)
0	42.7	76	51.06
2	42.92	78	51.28
4	43.14	80	51.5
6	43.36	82	51.72
8	43.58	84	51.94
10	43.8	86	52.16
12	44.02	88	52.38
14	44.24	90	52.6
16	44.46	92	52.82
18	44.68	94	53.04
20	44.9	96	53.26
22	45.12	98	53.48
24	45.34	100	53.7
26	45.56	102	53.92
continued...			

Power (W)	T _{CASE_MAX} (°C)		Power (W)	T _{CASE_MAX} (°C)
28	45.78		104	54.14
30	46		106	54.36
32	46.22		108	54.58
34	46.44		110	54.8
36	46.66		112	55.02
38	46.88		114	55.24
40	47.1		116	55.46
42	47.32		118	55.68
44	47.54		120	55.9
46	47.76		122	56.12
48	47.98		124	56.34
50	48.2		126	56.56
52	48.42		128	56.78
54	48.64		130	57
56	48.86		132	57.22
58	49.08		134	57.44
60	49.3		136	57.66
62	49.52		138	57.88
64	49.74		140	58.1
66	49.96		142	58.32
68	50.18		144	58.54
70	50.4		146	58.76
72	50.62		148	58.98
74	50.84		150	59.2